potential of machine learning in automating historically manual processes, thereby reducing the workload on insurance assessors and enhancing fairness in policy pricing. For customers, the system can act as an empowering tool that offers transparency regarding estimated premium values, leading to informed financial planning and better decision-making. Future work may involve incorporating additional features such as lifestyle habits, medical history, income levels, and real-time health monitoring data to further enhance predictive accuracy. The model can also be extended into a fully functional digital service through mobile or web applications, enabling widespread accessibility. Overall, the project successfully establishes that machine learning is not only capable of accurately predicting medical insurance premiums but also holds immense potential to modernize and streamline the insurance industry as a whole.

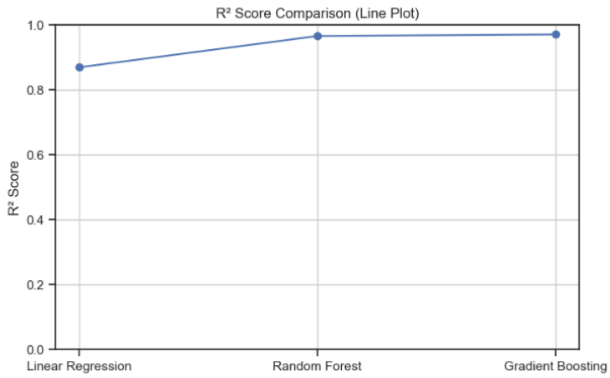

Fig 6: R Squared Values

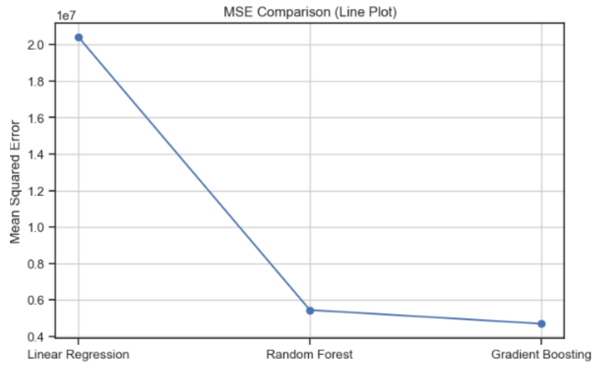

Fig 7: MSE Values

Dataset Taken From : Kaggle - https://www.kaggle.com/datasets/mirichoi0218/insurance

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